Comparative Effectiveness of Glucagon-Like Peptide-1 Receptor Agonists and Sodium/Glucose Cotransporter 2 Inhibitors in Preventing Chronic Kidney Failure and Mortality in Patients With Type 2 Diabetes and CKD

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Rationale & Objective: Glucagon-like peptide-1 receptor agonists (GLP-1RAs) and sodium/glucose cotransporter 2 (SGLT2) inhibitors improve cardiovascular, kidney, and survival outcomes in patients with type 2 diabetes; however, the comparative effectiveness of these drugs in a real-world setting remains unclear.

Study Design: Retrospective cohort study.

Setting & Participants: 79,047 patients with type 2 diabetes and an estimated glomerular filtration rate <60 mL/min/1.73 m^2 in the Taiwan National Health Insurance Research Database between 2016 and 2021.

Exposure: Treatment with GLP-1RAs or SGLT2 inhibitors.

Outcome: Initiation of kidney replacement therapy (KRT) and all-cause mortality.

Analytic Approach: Propensity score matching was performed to balance baseline characteristics between the groups. Cox proportional hazards models were used to estimate HRs and 95% CIs for each outcome using an intention-to-treat approach.

Results: 14,182 (7,091 initiating GLP-1RAs and 7,091 initiating SGLT2 inhibitors) individuals among the original cohort of 79,047 were included in the propensity score–matched analysis. With a median follow-up duration of 2.5 years, people initiating GLP-1RAs had a higher risk of requiring KRT than those initiating SGLT2 inhibitors (HR, 1.39; 95% CI, 1.19-1.63). Although tests of interaction did not have statistically significant findings, stratified analyses suggested possibly greater differences between the 2 drugs among patients with an estimated glomerular filtration rate <45 mL/min/1.73 m^2 or a urine albumin-creatinine ratio >300 mg/g. Overall mortality did not differ between treatment groups.

Limitations: Nonrandomized treatment selection.

Conclusions: Patients receiving SGLT2 inhibitors exhibited lower rates of progression to KRT than those receiving GLP-1RAs. These findings may inform the choice of these therapies in the setting of chronic kidney disease and type 2 diabetes.

Visual Abstract online

Complete author and article information provided before references.

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Chronic kidney disease (CKD) affects approximately 40% of people with type 2 diabetes[1] and is associated with significantly increased risks of kidney failure, cardiovascular events, and mortality.[2] Clinical trials have demonstrated that glucagon-like peptide-1 receptor agonists (GLP-1RAs) and sodium/glucose cotransporter-2 (SGLT2) inhibitors confer significant kidney protection and survival benefits.[3,4] The CREDENCE,[5] DAPA-CKD,[6] and EMPA-KIDNEY[7] trials documented the efficacy of SGLT2 inhibitors for reducing CKD progression. An earlier meta-analysis of trials reporting kidney outcomes in exploratory analyses suggested that GLP-1RAs may reduce composite kidney end points.[3] Additionally, in the recently published FLOW trial,[8] the long-acting GLP-1RA semaglutide significantly reduced the risk of clinically important kidney outcomes in patients with type 2 diabetes. However, no randomized trial has included a head-to-head comparison between these 2 drugs to evaluate their relative roles in preventing kidney failure and death among patients with diabetes and CKD, especially in patients with varying baseline kidney function, albuminuria severity, and comorbidities such as cardiovascular disease or obesity.

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Previous studies have yielded mixed results regarding the comparative effectiveness of SGLT2 inhibitors and GLP-1RAs on the risk of kidney outcomes. Network meta-analyses have included randomized trials comparing SGLT2 inhibitors or GLP-1RAs versus placebo, standard care, or other glucose-lowering treatment in adults with type 2 diabetes, and indirect comparisons suggest the superiority of SGLT2 inhibitors over GLP-1RAs.[9-11] Observational studies have produced inconsistent results, with some indicating no significant difference between these 2 drug classes[12-14] and others suggesting better kidney outcomes with SGLT2 inhibitors.[15-18] These discrepancies are probably due to diverse definitions of kidney outcomes, varying follow-up periods, and different baseline characteristics of the patient populations.

In the present study, we aimed to assess the comparative effectiveness of GLP-1RAs and SGLT2 inhibitors with regard to progression to kidney failure, including the need for kidney replacement therapy (KRT), and mortality among patients with type 2 diabetes and CKD. Our focus on a high-risk population enabled us to evaluate the

PLAIN-LANGUAGE SUMMARY

Chronic kidney disease is a major complication of type 2 diabetes that often leads to kidney failure and increased mortality. This study aimed to compare the effectiveness of 2 drug classes, glucagon-like peptide-1 receptor agonists and sodium/glucose cotransporter 2 inhibitors, in preventing kidney failure and death in patients with type 2 diabetes and chronic kidney disease. Using a nationwide health database in Taiwan, we applied rigorous statistical methods to balance differences between treatment groups and analyze outcomes. Our findings demonstrated that sodium/glucose cotransporter 2 inhibitors might be more effective than glucagon-like peptide-1 receptor agonists in reducing the risk of kidney failure, and possibly even more so in patients with advanced kidney disease. These results may inform the choice of these agents in the setting of chronic kidney disease and diabetes.

relative treatment effects in a susceptible population and identify the patients who are more likely to benefit from a particular drug treatment.

Methods

Data Source

This study was conducted using data from the Taiwan National Health Insurance Research Database, which includes demographic and enrollment records, diagnoses, procedures, and pharmacy dispensing claims for approximately 99% of Taiwan's population.[19] Since 2015, the database has been expanded to include laboratory results, blood pressure, and anthropometric measurements.[20] The present study protocol was approved by the National Taiwan University Research Ethics Committee, and the requirement of informed consent was waived because it was a retrospective analysis of deidentified data. The methods adhered to the Strengthening the Reporting of Observational Studies in Epidemiology guidelines.

Study Population and Study Drugs

Our retrospective cohort study included patients with type 2 diabetes, aged 20-100 years, with an estimated glomerular filtration rate (eGFR) of <60 mL/min/1.73 m^2 (ie, CKD stage G3a-G5) who started treatment with a GLP-1RA (liraglutide, dulaglutide, or semaglutide) or SGLT2 inhibitor (canagliflozin, dapagliflozin, empagliflozin, or ertugliflozin) between May 1, 2016 (when the first SGLT2 inhibitor was reimbursed by Taiwan's National Health Insurance), and December 31, 2021. The codes of study drugs are presented in Table S1.

The study employed a new user design in which we compared patients who received initial treatment with a GLP-1RA versus an SGLT2 inhibitor. The index date of enrollment in the cohort was defined by the first filled prescription for the study drug, with the requirement that patients had not received any prescription for the comparator drugs on the index date or during the preceding 12 months. We assessed eGFR using the CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) equation[21] based on the most recent outpatient measurements acquired within 1 year before the patient's index date.

Inclusion criteria were continuous National Health Insurance coverage for ≥1 year, diagnosis of type 2 diabetes (*International Classification of Diseases*, *9th Revision*, *Clinical Modification* code 250 or *International Classification of Diseases*, *10th Revision*, *Clinical Modification* code E08, E09, E10, E11, or E13), and at least one baseline eGFR laboratory result within 12 months before cohort entry. Exclusion criteria were diagnosis of type 1 diabetes and any high-risk conditions outlined in kidney protection trials, such as HIV, polycystic kidney disease, nephritis, nephrotic syndrome, organ transplant, and history of dialysis before cohort entry. Diagnostic codes are presented in Table S2.

Outcomes and Follow-up

The primary outcome was progression to KRT, defined by the issuance of a catastrophic illness certificate for chronic dialysis, initiation of long-term maintenance dialysis, or kidney transplant, as detailed in Table S3. We defined KRT based on a period of 90 days because all patients receiving dialysis for >90 days in Taiwan can apply for dialysis-related catastrophic cards in the National Health Insurance program. Patients with catastrophic illness certification who receive care for the illness or related conditions within the certificate's validity period do not need to pay a copayment for outpatient or inpatient care. Additionally, we analyzed a composite kidney outcome of KRT or a 40% decrease in eGFR to capture a broader range of clinically significant events. As the secondary outcome, we assessed overall mortality rates by linking patient records to the National Death Registry using unique identification numbers.

Adherence was assessed by analyzing discontinuation and therapy switching rates during the first 3 years of follow-up. Given the suboptimal adherence to treatment in the real world, we applied an intention-to-treat approach to analyze participants according to their initial treatment assignment, and provided more conservative comparative effect estimates. Participants were followed from the index date until the earliest of one of the following: documentation of KRT, disenrollment, study end (December 31, 2021), or death. Follow-up was further censored at 1, 2, and 3 years to focus on short- and medium-term effects. This approach aligned with pivotal trials using similar follow-up periods, ensuring comparability with existing evidence.[5,6,8]

Covariate Assessment

We retrieved information about patient characteristics from 1 year before their index date, capturing a broad array of data, including demographic characteristics, health conditions, medication use, and health care services use. To better adjust for patient complexity and comorbidity burden, we integrated the diabetes complication severity index,[22] Charlson comorbidity index,[23] and a claims-based frailty index.[24] With this comprehensive data collection, we aimed to capture potential outcome predictors (detailed in Tables S4 and S5). We also considered five key clinical parameters—glycated hemoglobin (ie, hemoglobin A_{1c} [HbA_{1c}]), serum creatinine, low-density lipoprotein cholesterol, urine albumin-creatinine ratio (UACR), and systolic blood pressure—measured within 365 days before treatment initiation, selecting the measurement closest to the index date for analysis. These variables were analyzed as categorical variables. Grouping details and a full list of covariates are provided in Table 1.

Statistical Analysis

Baseline characteristics of the 2 treatment groups are presented as means and standard deviations for continuous variables and as frequencies and proportions for categorical variables. We used the missing-indicator method to handle missing data regarding clinical parameters.[25] Propensity scores (PSs) were estimated via logistic regression, incorporating the covariates described above (Table 1), and matched using a nearest-neighbor method with a caliper of 0.025 on the PS scale. The matching performance was visually assessed through PS distribution plots and quantitatively assessed by standardized differences, with values lower than 0.1 indicating a successful match.[26]

In the auxiliary analysis, overlap weighting was applied to balance baseline characteristics between treatment groups and ensure valid comparisons within the region of clinical equipoise.[27-29] All subjects from the original cohort were included, but their contributions varied based on PSs. Subjects near clinical equipoise (scores close to 0.5) were assigned higher weights, whereas those at the extremes contributed minimally. Individuals in nonoverlapping regions received zero weights, effectively excluding them from the analysis. This approach emphasized clinically relevant comparisons while achieving covariate balance.

We calculated the event counts, follow-up durations, and incidence rates per 1,000 person-years. Hazard ratios (HRs) and 95% confidence intervals (CIs) were estimated by using a Cox proportional hazards models in the original, PS-matched, and overlap-weighted cohorts. We generated cumulative incidence curves using the Kaplan-Meier method to illustrate time-to-event data and validated the proportional hazards assumptions using Schoenfeld residuals.

Sensitivity and Subgroup Analyses

To ensure the robustness of our findings, we conducted several sensitivity analyses. First, patients with conditions such as cancer and autoimmune diseases (codes detailed in Table S2) were excluded to reduce residual confounding associated with significant variability in kidney function decline rate. Second, we excluded patients with severe kidney impairment (eGFR <15 mL/min/1.73 m^2) because of contraindications to SGLT2 inhibitor use. Third, multiple imputation was applied to address missing HbA_{1c} and UACR data in the PS-matched cohort, mitigating biases from incomplete datasets.[30] Finally, PS matching was refined by using a logit scale with a caliper width of 0.2 times the standard deviation of the logit of the PS, improving the precision of matching and reducing residual imbalances.[31]

We also conducted prespecified subgroup analyses stratified according to baseline eGFR categories (45-59 [stage G3a] and <45 mL/min/1.73 m^2 [stage G3b-G5]) and UACR levels (≥300 [stage A3] and <300 mg/g [stage A1/A2]) to explore potential treatment effect modifications. For each subgroup, PSs were recalculated and rematched or weighted. Interactions were assessed by incorporating product terms between treatment and covariates into the regression model applied to the PS-matched and overlap-weighted cohorts. All analyses were executed using SAS software, version 9.3, and statistical significance was defined by a P value of less than 0.05.

Results

Study Population

Among nearly 2 million adults with type 2 diabetes, we identified a cohort of 79,047 individuals, including 8,039 initiating GLP-1RA therapy (47% liraglutide, 50% dulaglutide, and 3% semaglutide) and 71,008 initiating SGLT2 inhibitor therapy (40% dapagliflozin, 10% canagliflozin, and 49% empagliflozin; Fig S1). Individuals initiating GLP-1RA therapy were generally younger, included a lower proportion of men (53% vs 57%), and were more likely to be overweight or obese compared with those initiating SGLT2 inhibitor therapy. They also exhibited worse kidney function, as indicated by lower eGFR and higher UACR, along with diabetes complications of greater severity and higher HbA_{1c} levels (Table 1 and Table S6). Individuals initiating GLP-1RA treatment had higher Charlson comorbidity indexes, more frequent episodes of acute kidney injury, and a greater incidence of hyperkalemia or potassium binder use. Additionally, they were more likely to use insulin and had greater numbers of antidiabetic and cardiovascular drug prescriptions. Regarding health care use, GLP-1RA users required more overall and cardiovascular-related outpatient visits. Adherence patterns showed that people using GLP-1RAs had higher discontinuation and switching rates than those using SGLT2 inhibitors, particularly in the first year (32.9% vs 28.3% for discontinuation; 5.29% vs 1.39% for switching), with rates declining substantially thereafter (Table S7).

After PS matching, the analysis included 14,182 patients, achieving balanced group characteristics, with standardized differences of less than 0.1 (Table 1) and fully overlapping PS distributions (Fig S2). Within this group, the average age was 67.4 years, 54% were male, and 70% had ever received insulin therapy. At baseline, 32% had

Table 1. Baseline Demographic Characteristics, Comorbidities, Other Medication Use, and Resource Use Measured Within 365 Days Before the Index Date Before and After Propensity Score Matching

Variable	Original Cohort (n = 79,047)			Matched Cohort (n = 14,182)		
	GLP-1RA (n = 8,039)	SGLT2 Inhibitor (n = 71,008)	Standardized Difference	GLP-1RA (n = 7,091)	SGLT2 Inhibitor (n = 7,091)	Standardized Difference
Patient Demographic Characteristics						
Age at diabetes diagnosis, y	67.22 ± 11.38	69.53 ± 10.56	−0.210	67.53 ± 11.25	67.33 ± 11.36	0.018
Male sex	4,264 (53.04%)	40,729 (57.36%)	−0.087	3,802 (53.62%)	3,802 (53.62%)	<0.001
Clinical Parameters						
eGFR group						
45-59 mL/min/1.73 m^2	2,606 (32.42%)	43,981 (61.94%)	−0.619	2,589 (36.51%)	2,633 (37.13%)	−0.013
30-44 mL/min/1.73 m^2	2,819 (35.07%)	21,044 (29.64%)	0.116	2,622 (36.98%)	2,621 (36.96%)	<0.001
15-29 mL/min/1.73 m^2	2,134 (26.55%)	5,060 (7.13%)	0.537	1,554 (21.92%)	1,526 (21.52%)	0.010
<15 mL/min/1.73 m^2	480 (5.97%)	923 (1.30%)	0.252	326 (4.60%)	311 (4.39%)	0.010
HbA_{1c} group						
>9.0%	3,892 (48.41%)	17,172 (24.18%)	0.521	3,280 (46.26%)	3,361 (47.40%)	−0.023
7.0-9.0%	3,358 (41.77%)	34,655 (48.80%)	−0.142	3,063 (43.20%)	3,021 (42.60%)	0.012
<7.0%	711 (8.84%)	17,267 (24.32%)	−0.425	670 (9.45%)	634 (8.94%)	0.018
Missing	78 (0.97%)	1,914 (2.70%)	−0.129	78 (1.10%)	75 (1.06%)	0.004
LDL group						
>140 mg/dL	610 (7.59%)	5,185 (7.30%)	0.011	545 (7.69%)	575 (8.11%)	−0.016
120-140 mg/dL	702 (8.73%)	6,364 (8.96%)	−0.008	619 (8.73%)	610 (8.60%)	0.005
100-119 mg/dL	1,280 (15.92%)	11,017 (15.52%)	0.011	1,123 (15.84%)	1,088 (15.34%)	0.014
<100 mg/dL	5,057 (62.91%)	43,188 (60.82%)	0.043	4,436 (62.56%)	4,432 (62.50%)	0.001
Missing	390 (4.85%)	5,254 (7.40%)	−0.106	368 (5.19%)	386 (5.44%)	−0.011
UACR group						
<30 mg/g	744 (9.25%)	9,689 (13.64%)	−0.138	706 (9.96%)	717 (10.11%)	−0.005
30 to <300 mg/g	1,255 (15.61%)	11,343 (15.97%)	−0.010	1,152 (16.25%)	1,080 (15.23%)	0.028
>300 mg/g	2,442 (30.38%)	17,038 (23.99%)	0.144	2,090 (29.47%)	2,136 (30.12%)	−0.014
Missing	3,598 (44.76%)	32,938 (46.39%)	−0.033	3,143 (44.32%)	3,158 (44.54%)	−0.004
Diabetes Complications						
Nephropathy	5,921 (73.65%)	42,401 (59.71%)	0.299	5,085 (71.71%)	5,009 (70.64%)	0.024
Retinopathy	440 (5.47%)	1,962 (2.76%)	0.137	362 (5.11%)	364 (5.13%)	−0.001
Neuropathy	1,572 (19.55%)	10,021 (14.11%)	0.146	1,355 (19.11%)	1,415 (19.95%)	−0.021
Diabetes complication severity index	2.61 ± 1.72	2.20 ± 1.62	0.247	2.55 ± 1.72	2.55 ± 1.72	<−0.001
Comorbidities						
Overweight or obesity	399 (4.96%)	1,324 (1.86%)	0.171	311 (4.39%)	339 (4.78%)	−0.019

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Table 1 (Cont'd). Baseline Demographic Characteristics, Comorbidities, Other Medication Use, and Resource Use Measured Within 365 Days Before the Index Date Before and After Propensity Score Matching

Variable	Original Cohort (n = 79,047)			Matched Cohort (n = 14,182)		
	GLP-1RA (n = 8,039)	SGLT2 Inhibitor (n = 71,008)	Standardized Difference	GLP-1RA (n = 7,091)	SGLT2 Inhibitor (n = 7,091)	Standardized Difference
Hypertension	6,976 (86.78%)	59,947 (84.42%)	0.067	6,127 (86.41%)	6,123 (86.35%)	0.002
Ischemic heart disease	2,519 (31.33%)	23,944 (33.72%)	−0.051	2,259 (31.86%)	2,298 (32.41%)	−0.012
Cerebrovascular disease	1,531 (19.04%)	12,678 (17.85%)	0.031	1,357 (19.14%)	1,390 (19.60%)	−0.012
Cardiac dysrhythmia	869 (10.81%)	9,949 (14.01%)	−0.097	787 (11.10%)	785 (11.07%)	<0.001
Congestive heart failure	1,378 (17.14%)	12,753 (17.96%)	−0.022	1,216 (17.15%)	1,204 (16.98%)	0.004
Peripheral vascular disease	489 (6.08%)	3,439 (4.84%)	0.055	412 (5.81%)	424 (5.98%)	−0.007
Hyperlipidemia	6,224 (77.42%)	53,186 (74.90%)	0.059	5,480 (77.28%)	5,500 (77.56%)	−0.007
Chronic lung disease	1,309 (16.28%)	11,714 (16.50%)	−0.006	1,183 (16.68%)	1,181 (16.65%)	<0.001
Edema	754 (9.38%)	5,616 (7.91%)	0.052	644 (9.08%)	634 (8.94%)	0.005
Hyperkalemia diagnosis or potassium binder use	777 (9.67%)	3,065 (4.32%)	0.211	619 (8.73%)	615 (8.67%)	0.002
Acute kidney injury	682 (8.48%)	3,783 (5.33%)	0.125	593 (8.36%)	597 (8.42%)	−0.002
Genitourinary tract infection–related episodes	2,017 (25.09%)	15,081 (21.24%)	0.091	1,765 (24.89%)	1,858 (26.20%)	−0.030
Chronic liver disease	758 (9.43%)	7,435 (10.47%)	−0.035	688 (9.70%)	712 (10.04%)	−0.011
Gout	2,011 (25.02%)	15,602 (21.97%)	0.072	1,738 (24.51%)	1,681 (23.71%)	0.019
Cancer	666 (8.28%)	6,165 (8.68%)	−0.014	596 (8.41%)	592 (8.35%)	0.002
Autoimmune disease	328 (4.08%)	2,830 (3.99%)	0.005	286 (4.03%)	288 (4.06%)	−0.001
No. of cardiovascular-related diseases[a]	2.38 ± 1.04	2.34 ± 1.04	0.039	2.38 ± 1.05	2.39 ± 1.05	−0.012
Charlson comorbidity index	4.92 ± 2.16	4.36 ± 2.29	0.253	4.87 ± 2.20	4.86 ± 2.28	0.003
Claims-based frailty index	0.13 ± 0.13	0.14 ± 0.13	−0.097	0.13 ± 0.13	0.13 ± 0.13	0.005
Medication Use						
Antihyperglycemic medication use						
Any insulin	5,862 (72.92%)	24,404 (34.37%)	0.838	4,929 (69.51%)	4,981 (70.24%)	−0.016
Metformin	5,092 (63.34%)	59,767 (84.17%)	−0.487	4,844 (68.31%)	4,913 (69.29%)	−0.021
Sulfonylureas	5,130 (63.81%)	44,259 (62.33%)	0.031	4,605 (64.94%)	4,622 (65.18%)	−0.005
Glinides	2,026 (25.20%)	8,217 (11.57%)	0.357	1,625 (22.92%)	1,682 (23.72%)	−0.019
Pioglitazone	2,310 (28.73%)	17,113 (24.10%)	0.105	2,007 (28.30%)	2,006 (28.29%)	<0.001
α-Glucosidase inhibitors	2,331 (29.00%)	13,960 (19.66%)	0.219	2,023 (28.53%)	2,059 (29.04%)	−0.011
Dipeptidyl peptidase-4 inhibitors	6,612 (82.25%)	52,212 (73.53%)	0.211	5,788 (81.62%)	5,797 (81.75%)	−0.003
No. of oral antihyperglycemic medications[b]	2.92 ± 1.26	2.75 ± 1.17	0.139	2.95 ± 1.26	2.97 ± 1.25	−0.021

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Table 1 (Cont'd). Baseline Demographic Characteristics, Comorbidities, Other Medication Use, and Resource Use Measured Within 365 Days Before the Index Date Before and After Propensity Score Matching

Variable	Original Cohort (n = 79,047) GLP-1RA (n = 8,039)	Original Cohort SGLT2 Inhibitor (n = 71,008)	Original Cohort Standardized Difference	Matched Cohort (n = 14,182) GLP-1RA (n = 7,091)	Matched Cohort SGLT2 Inhibitor (n = 7,091)	Matched Cohort Standardized Difference
Non-antihyperglycemic medication use						
ACEI/ARB	6,634 (82.52%)	57,436 (80.89%)	0.042	5,845 (82.43%)	5,813 (81.98%)	0.012
Diuretics	4,411 (54.87%)	34,341 (48.36%)	0.131	3,823 (53.91%)	3,806 (53.67%)	0.005
Aldosterone antagonists and other potassium-sparing agents	944 (11.74%)	9,259 (13.04%)	−0.039	841 (11.86%)	864 (12.18%)	−0.010
Other antihypertensive agents	1,353 (16.83%)	9,050 (12.75%)	0.115	1,143 (16.12%)	1,165 (16.43%)	−0.008
Antiarrhythmic agents	555 (6.90%)	5,780 (8.14%)	−0.047	493 (6.95%)	509 (7.18%)	−0.009
Aspirin	3,448 (42.89%)	30,585 (43.07%)	−0.004	3,079 (43.42%)	3,086 (43.52%)	−0.002
Clopidogrel	1,204 (14.98%)	10,266 (14.46%)	0.015	1,077 (15.19%)	1,120 (15.79%)	−0.017
New oral anticoagulant	346 (4.30%)	4,687 (6.60%)	−0.101	322 (4.54%)	320 (4.51%)	0.001
Statins	6,555 (81.54%)	54,498 (76.75%)	0.118	5,740 (80.95%)	5,709 (80.51%)	0.011
Fibrates	1,594 (19.83%)	10,870 (15.31%)	0.119	1,363 (19.22%)	1,367 (19.28%)	−0.001
Urate-lowering agents	2,623 (32.63%)	18,890 (26.60%)	0.132	2,212 (31.19%)	2,175 (30.67%)	0.011
Febuxostat	1,645 (20.46%)	8,480 (11.94%)	0.233	1,309 (18.46%)	1,278 (18.02%)	0.011
Pentoxifylline	2,037 (25.34%)	12,470 (17.56%)	0.190	1,651 (23.28%)	1,623 (22.89%)	0.009
Antibiotics	5,279 (65.67%)	43,172 (60.80%)	0.101	4,637 (65.39%)	4,652 (65.60%)	−0.004
COX-2–selective NSAID	1,298 (16.15%)	13,389 (18.86%)	−0.071	1,182 (16.67%)	1,214 (17.12%)	−0.012
COX-2–nonselective NSAID	4,968 (61.80%)	44,195 (62.24%)	−0.009	4,392 (61.94%)	4,376 (61.71%)	0.005
No. of cardiovascular-related medications[c]	4.69 ± 1.98	4.44 ± 2.00	0.125	4.66 ± 1.99	4.65 ± 2.04	0.003
Access of Health Services						
Abdominal ultrasound	1,837 (22.85%)	15,141 (21.32%)	0.037	1,628 (22.96%)	1,652 (23.30%)	−0.008
HbA_{1c} measurement	8,018 (99.74%)	70,448 (99.21%)	0.073	7,070 (99.70%)	7,073 (99.75%)	−0.008
No. of hospitalizations	0.60 ± 1.06	0.51 ± 1.07	0.081	0.60 ± 1.09	0.62 ± 1.10	−0.019
For cardiovascular-related episodes[a]	0.51 ± 0.97	0.44 ± 0.96	0.076	0.52 ± 1.00	0.54 ± 1.00	−0.020
For hyperkalemia	0.02 ± 0.17	0.01 ± 0.13	0.075	0.02 ± 0.17	0.02 ± 0.18	−0.002
For acute kidney injury	0.07 ± 0.29	0.05 ± 0.23	0.102	0.07 ± 0.30	0.08 ± 0.29	−0.010
For genitourinary infection–related episodes[d]	0.12 ± 0.44	0.08 ± 0.36	0.097	0.12 ± 0.45	0.13 ± 0.45	−0.012
No. of outpatient visits	42.83 ± 24.13	39.61 ± 23.09	0.136	42.36 ± 23.94	42.19 ± 24.41	0.007
For cardiovascular-related episodes[a]	19.22 ± 11.38	17.10 ± 10.35	0.195	18.87 ± 11.13	18.83 ± 12.03	0.003

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Table 1 (Cont'd). Baseline Demographic Characteristics, Comorbidities, Other Medication Use, and Resource Use Measured Within 365 Days Before the Index Date Before and After Propensity Score Matching

Variable	Original Cohort (n = 79,047)			Matched Cohort (n = 14,182)		
	GLP-1RA (n = 8,039)	SGLT2 Inhibitor (n = 71,008)	Standardized Difference	GLP-1RA (n = 7,091)	SGLT2 Inhibitor (n = 7,091)	Standardized Difference
For genitourinary infection–related episodes[d]	1.16 ± 3.73	0.89 ± 3.18	0.076	1.12 ± 3.64	1.17 ± 3.89	−0.014

Abbreviations: ACEI, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; COX, cyclooxygenase; eGFR, estimated glomerular filtration rate; GLP-1RA, glucagon-like peptide-1 receptor agonist; LDL, low-density lipoprotein; SGLT2, sodium/glucose cotransporter 2; HbA_{1C}, glycated hemoglobin; NSAID, nonsteroidal anti-inflammatory drug; UACR, urine albumin-creatinine ratio.
Data presented as mean ± standard deviation where applicable.
[a]Cardiovascular-related diseases included hypertension, ischemic heart disease, cerebrovascular disease, congestive heart failure, peripheral vascular disease, and disorders of lipid metabolism.
[b]Oral antidiabetic medications included metformin, sulfonylureas, glinides, pioglitazone, α-glucosidase inhibitors, and dipeptidyl peptidase-4 inhibitors.
[c]Cardiovascular-related medications included ACEIs/ARBs, β-blocking agents, calcium channel blockers, diuretics, other antihypertensive agents, nitrates, antiarrhythmic agents, digoxin, aspirin, clopidogrel, warfarin, statins, and fibrates.
[d]Genitourinary infection–related episodes included urinary tract infection, genital tract infection, Fournier's gangrene, and candidiasis.

ischemic heart disease, 19% had cerebrovascular disease, and 17% had congestive heart failure. Laboratory data indicated an average HbA_{1c} level of 9.2% and eGFR of 40 mL/min/1.73 m^2.

Risks of Progression to KRT Associated With GLP-1RAs Versus SGLT2 Inhibitors

During a median follow-up of 2.5 years, progression to KRT occurred in 262 individuals (3.7%) initiating GLP-1RAs and 174 (2.5%) of those initiating SGLT2 inhibitors in the PS-matched cohort, corresponding to incidence rates of 17.7 and 11.9 per 1,000 person-years, respectively (Table 2). Compared with SGLT2 inhibitors, GLP-1RAs were associated with a universally higher risk of requiring KRT, with an adjusted HR of 1.39 (95% CI, 1.19-1.63), although the HR magnitude decreased over time to 1.72 (95% CI, 1.11-2.68) at 1 year, 1.73 (95% CI, 1.34-2.23) at 2 years, and 1.48 (95% CI, 1.22-1.79) at 3 years (Table 2).

Sensitivity analyses excluding patients with cancer, autoimmune diseases, or severe kidney impairment (eGFR <15 mL/min/1.73 m^2), using multiple imputation to handle missing data, and matching on the logit scale also supported the primary results (Table 3; detailed in Tables S8 and S9). Auxiliary analyses using a composite kidney end point and overlap weighting also demonstrated an increased risk of requiring KRT among GLP-1RA initiators versus SGLT2 inhibitor initiators (Table 3 and Table S10; detailed in Tables S11-S14). Schoenfeld residual tests confirmed that the proportional hazards assumption was met. Figure S3 depicts the cumulative incidence of progression to KRT over time in the PS-matched and overlap-weighted cohorts.

Risks of Overall Mortality Associated with GLP-1RAs Versus SGLT2 Inhibitors

In PS-matched and overlap-weighted analyses, over 3 years of follow-up, individuals initiating GLP-1RAs and SGLT2 inhibitors exhibited comparable risks for overall mortality, with HRs of 0.94 (95% CI, 0.86-1.03) and 0.94 (95% CI, 0.87-1.03), respectively (Tables 2 and 3). The cumulative incidence plots in Fig S4 illustrate how the probability of death evolved over time in both treatment groups.

Subgroup Analyses

In the subgroup analysis, individuals initiating GLP-1RAs had a higher risk of progressing to KRT than those initiating SGLT2 inhibitors, particularly among those with an eGFR <45 mL/min/1.73 m^2 (HR, 1.56, 95% CI, 1.31-1.86 in PS-matched analysis; HR, 1.44; 95% CI, 1.23-1.68 in overlap-weighted analysis) and UACR ≥300 mg/g (HR, 1.33; 95% CI, 1.03-1.71 in PS-matched; HR, 1.33; 95% CI, 1.06-1.67 in overlap-weighted analysis; Table 4 and Table S14). Risks were comparable for eGFR ≥45 mL/min/1.73 m^2 or UACR <300 mg/g. Although the results of interaction tests were not significant, these findings suggest a potentially stronger impact of GLP-1RAs in advanced kidney disease, warranting further investigation.

Table 2. Follow-up Duration, Incidence Rates, and Hazard Ratios of Outcomes for GLP-1RA versus SGLT2 Inhibitor Initiators in Original and Propensity Score–Matched Cohorts

	Original Cohort (n = 79,047)		Matched Cohort (n = 14,182)	
	GLP-1RA (n = 8,039)	SGLT2 Inhibitor (n = 71,008)	GLP-1RA (n = 7,091)	SGLT2 Inhibitor (n = 7,091)
Intention-to-Treat Approach, Censored at 1y				
Kidney replacement therapy				
Total follow-up PY	7,242	58,351	6,341	6,273
Follow-up year				
Median	1.00	1.00	1.00	1.00
IQR	1.00-1.00	0.73-1.00	1.00-1.00	1.00-1.00
Number of cases	88	88	54	31
Incidence rate per 1,000 PY (95% CI)	12.15 (9.86-14.97)	1.51 (1.22-1.86)	8.52 (6.52-11.12)	4.94 (3.48-7.03)
Hazard ratio (95% CI)	7.81 (5.81-10.49)		1.72 (1.11-2.68)	
Overall mortality				
Total follow-up PY	7,267	58,379	6,355	6,282
Follow-up year				
Median	1.00	1.00	1.00	1.00
IQR	1.00-1.00	0.73-1.00	1.00-1.00	1.00-1.00
Number of deaths	360	2,741	312	346
Incidence rate per 1,000 PY (95% CI)	49.5 (44.7-54.9)	47.0 (45.2-48.7)	49.1 (43.9-54.9)	55.1 (49.6-61.2)
Hazard ratio (95% CI)	1.07 (0.96-1.19)		0.89 (0.77-1.04)	
Intention-to-Treat Approach, Censored at 2y				
Kidney replacement therapy				
Total follow-up PY	12,961	96,150	11,308	11,161
Follow-up year				
Median	2.00	1.61	2.00	2.00
IQR	1.27-2.00	0.73-2.00	1.21-2.00	1.15-2.00
Number of cases	266	277	165	94
Incidence rate per 1,000 PY (95% CI)	20.5 (18.2-23.1)	2.88 (2.56-3.24)	14.6 (12.5-17.0)	8.42 (6.88-10.3)
Hazard ratio (95% CI)	6.64 (5.61-7.86)		1.73 (1.34-2.23)	
Overall mortality				
Total follow-up PY	13,130	96,289	11,406	11,214
Follow-up year				
Median	2.00	1.61	2.00	2.00
IQR	1.35-2.00	0.73-2.00	1.25-2.00	1.17-2.00
Number of deaths	623	4,085	533	559
Incidence rate per 1,000 PY (95% CI)	47.5 (43.9-51.3)	42.4 (41.1-43.8)	46.7 (42.9-50.8)	49.9 (45.9-54.2)
Hazard ratio (95% CI)	1.14 (1.05-1.24)		0.94 (0.83-1.06)	
Intention-to-Treat Approach, Censored at 3y				
Kidney replacement therapy				
Total follow-up PY	16,993	119,647	14,812	14,585
Follow-up year				
Median	2.51	1.61	2.48	2.40
IQR	1.27-3.00	0.73-3.00	1.21-3.00	1.15-3.00
Number of cases	422	468	262	174
Incidence rate per 1,000 PY (95% CI)	24.8 (22.6-27.3)	3.91 (3.57-4.28)	17.7 (15.7-20.0)	11.9 (10.3-13.8)
Hazard ratio (95% CI)	5.83 (5.11-6.65)		1.48 (1.22-1.79)	
Overall mortality				
Total follow-up PY	17,380	119,989	15,035	14,718
Follow-up year				
Median	2.61	1.61	2.55	2.45
IQR	1.35-3.00	0.73-3.00	1.25-3.00	1.17-3.00
Number of deaths	839	5,024	710	721
Incidence rate per 1,000 PY (95% CI)	48.3 (45.1-51.7)	41.9 (40.7-43.0)	47.2 (43.9-50.8)	49.0 (45.5-52.7)
Hazard ratio (95% CI)	1.17 (1.09-1.26)		0.96 (0.87-1.07)	

(Continued)

Table 2 (Cont'd). Follow-up Duration, Incidence Rates, and Hazard Ratios of Outcomes for GLP-1RA versus SGLT2 Inhibitor Initiators in Original and Propensity Score–Matched Cohorts

	Original Cohort (n = 79,047)		Matched Cohort (n = 14,182)	
	GLP-1RA (n = 8,039)	SGLT2 Inhibitor (n = 71,008)	GLP-1RA (n = 7,091)	SGLT2 Inhibitor (n = 7,091)
Full Time Analysis				
Kidney replacement therapy				
Total follow-up PY	20,210	140,617	17,729	17,605
Follow-up year				
Median	2.51	1.61	2.48	2.40
IQR	1.27-3.59	0.73-3.00	1.21-3.62	1.15-3.67
Number of cases	558	719	363	261
Incidence rate per 1,000 PY (95% CI)	27.6 (25.4-30.0)	5.11 (4.75-5.50)	20.5 (18.5-22.7)	14.8 (13.1-16.7)
Hazard ratio (95% CI)	5.05 (4.52-5.64)		1.39 (1.19-1.63)	
Overall mortality				
Total follow-up PY	20,934	141,417	18,175	17,914
Follow-up year				
Median	2.61	1.61	2.55	2.45
IQR	1.35-3.71	0.73-3.01	1.25-3.72	1.17-3.74
Number of deaths	1,025	5,994	860	903
Incidence rate per 1,000 PY (95% CI)	49.0 (46.1-52.1)	42.4 (41.3-43.5)	47.3 (44.3-50.6)	50.4 (47.2-53.8)
Hazard ratio (95% CI)	1.17 (1.10-1.25)		0.94 (0.86-1.03)	

Abbreviations: CI, confidence interval; GLP-1RA, glucagon-like peptide-1 receptor agonist; SGLT2, sodium/glucose cotransporter 2; PY, person-year.
Hazard ratios for the original cohort represent crude estimates from the pre–propensity score matching cohort, reflecting unadjusted treatment effects without baseline covariate adjustment. Adjusted hazard ratios were derived from the propensity score–matched cohort, whereby baseline differences between treatment groups were balanced through propensity score matching.

Discussion

Here we performed a comprehensive analysis of a nationwide cohort of patients with type 2 diabetes and eGFR <60 mL/min/1.73 m^2 in a real-world setting. Our results indicated that, compared with initial treatment with an SGLT2 inhibitor, initial treatment with a GLP-1RA was associated with a higher risk of progression to KRT, particularly in patients with an eGFR <45 mL/min/1.73 m^2 or with a severely increased UACR >300 mg/g. Mortality did not differ between the 2 treatment groups. These findings suggest that, although GLP-1RAs offer kidney-protective benefits, they may be less effective in preventing KRT progression compared with SGLT2 inhibitors, especially in patients with higher kidney risk profiles.

Previously, the EMPA-KIDNEY trial results highlighted that empagliflozin may have greater kidney benefit among patients with higher UACRs.[7] Those findings combined with the present results underscore the importance of personalized therapeutic decisions based on individual kidney profiles for the management of patients with type 2 diabetes and CKD.

Observational studies comparing the kidney outcomes associated with GLP-1RA versus SGLT2 inhibitor treatment have yielded varying results. Some prior evidence suggests that treatment effectiveness is influenced by baseline eGFR or albuminuria status. Xie et al conducted a study using Veterans Affairs databases and demonstrated that treatment with SGLT2 inhibitors and GLP-1RAs was associated with similar risks of composite kidney outcomes (HR, 0.95; 95% CI, 0.87-1.04).[12] Notably, their study revealed that SGLT2 inhibitor treatment was associated with a 20% lower risk (HR, 0.81; 95% CI, 0.57-1.16) in patients with an eGFR <45 mL/min/1.73 m^2. Edmonston et al analyzed data from 20 U.S. health systems contributing to PCORnet and found that kidney outcomes did not differ between SGLT2 inhibitor and GLP-1RA treatment among the overall study participants with type 2 diabetes (HR, 0.91; 95% CI, 0.81-1.02).[13] However, in patients with an eGFR <60 mL/min/1.73 m^2, SGLT2 inhibitors were associated with lower HRs of 0.68 (95% CI, 0.42-1.11) for incident end-stage kidney disease and 0.71 (95% CI, 0.43-1.17) for incident dialysis. Baseline UACR level did not significantly modify the association between treatment group and kidney outcome.[13] Another study analyzed the Hong Kong Hospital Authority database using an on-treatment approach and revealed that SGLT2 inhibitor users had a lower risk of composite kidney outcomes, mainly driven by reductions of end-stage kidney disease. The authors also reported lower HRs for patients with an eGFR <60 mL/min/1.73 m^2 and for patients with macroalbuminuria, although no significant interaction was found.[17] Whereas previous studies have included patients with a broad range of eGFR values (mostly 70-90 mL/min/1.73 m^2) and have investigated composite kidney outcomes, the present study focused on patients with type 2 diabetes and moderate to severe kidney disease, and specifically investigated the clinically important outcome of KRT.

Table 3. Sensitivity Analysis: Adjusted Hazard Ratios and 95% Confidence Intervals of Outcomes Comparing Those Who Initiated GLP-1RAs Versus SGLT2 Inhibitors

Outcome	PS-Matched Cohort Excluding Cancer or Autoimmune Disease (n = 12,520)	PS-Matched Cohort Excluding Those With eGFR <15 mL/min/1.73 m^2 (n = 13,534)	PS-Matched Cohort Using Multiple Imputation for Missing HbA$_{1c}$ and UACR Data[a]	PS-Matched With Logit Scale (n = 14,220)	Overlap-Weighted Cohort (n = 10,892)[b]
Intention-to-Treat Approach, Censored at 1 y					
KRT	1.52 (0.96-2.40)	2.82 (1.11-7.14)	1.72 (1.05-2.81)	1.72 (1.11-2.68)	1.92 (1.31-2.82)
Overall mortality	0.85 (0.72-1.01)	0.92 (0.78-1.08)	0.87 (0.75-1.02)	0.88 (0.76-1.03)	0.93 (0.82-1.06)
Intention-to-Treat Approach, Censored at 2 y					
KRT	1.54 (1.18-2.00)	1.79 (1.28-2.51)	1.61 (1.21-2.15)	1.74 (1.35-2.24)	1.77 (1.42-2.20)
Overall mortality	0.90 (0.79-1.03)	0.94 (0.83-1.07)	0.89 (0.78-1.02)	0.93 (0.83-1.05)	0.95 (0.86-1.05)
Intention-to-Treat Approach, Censored at 3 y					
KRT	1.42 (1.16-1.73)	1.46 (1.16-1.84)	1.46 (1.18-1.82)	1.48 (1.23-1.80)	1.55 (1.31-1.83)
Overall mortality	0.90 (0.81-1.01)	0.94 (0.84-1.04)	0.90 (0.80-1.01)	0.96 (0.87-1.06)	0.96 (0.88-1.05)
Full Time Analysis					
KRT	1.28 (1.09-1.52)	1.30 (1.09-1.56)	1.33 (1.11-1.60)	1.40 (1.19-1.64)	1.45 (1.26-1.67)
Overall mortality	0.88 (0.79-0.97)	0.93 (0.84-1.02)	0.88 (0.79-1.00)	0.94 (0.85-1.03)	0.94 (0.87-1.03)

Abbreviations: eGFR, estimated glomerular filtration rate; GLP-1RA, glucagon-like peptide-1 receptor agonist; HbA$_{1c}$, hemoglobin A$_{1c}$; KRT, kidney replacement therapy; PS, propensity score; SGLT2, sodium/glucose cotransporter 2.
[a]Missing data were handled using multiple imputation (20 datasets). Propensity scores were estimated and matched within each dataset. Cox models were fitted, and pooled hazard ratios were calculated using Rubin's rules.
[b]The case number in the overlap-weighted cohort represents the effective sample size, calculated as $(\sum\omega i)^2/\sum\omega i^2$. All subjects were included in the analysis, but individuals in nonoverlapping regions received zero weights, and those near clinical equipoise contributed more significantly.[29]

Our study observed a 39% higher hazard for KRT progression with GLP-1RAs compared with SGLT2 inhibitors, which was similar to the findings reported in network meta-analyses by Yamada et al[32] and Zhang et al,[33] but higher than that reported by Palmer et al.[9] Our study included patients with a lower obesity prevalence, advanced CKD, and poor glycemic control (mean HbA$_{1c}$, 9.2%). Additionally, higher discontinuation and switching rates among people using GLP-1RAs and the limited use of semaglutide, the only GLP-1RA with proven kidney-protective effects, may explain the reduced kidney benefits observed with GLP-1RAs. The use of KRT as a clinically significant end point, combined with the short follow-up period, likely highlighted the immediate benefits of SGLT2 inhibitors.

The kidney-protective effects of SGLT2 inhibitors and GLP-1RAs involve shared and distinct mechanisms. SGLT2 inhibitors act by blocking glucose and sodium

Table 4. Subgroup Analysis: Adjusted Hazard Ratios and 95% Confidence Intervals of Outcomes Comparing Propensity Score–Matched Cohort of Those Who Initiated GLP-1RAs Versus SGLT2 Inhibitors

	eGFR			UACR		
Outcome	45-59 mL/min/1.73 m^2 (n = 5,194)	<45 mL/min/1.73 m^2 (n = 8,840)	*P* Value[a]	<300 mg/g (n = 3,728)	≥300 mg/g (n = 4,138)	*P* Value[a]
Intention-to-Treat Approach, Censored at 1 y						
KRT	NA	1.93 (1.19-3.13)	NA	NA	2.67 (1.24-5.75)	NA
Overall mortality	0.97 (0.71-1.32)	0.87 (0.73-1.04)	0.5	0.70 (0.49-0.99)	0.91 (0.67-1.22)	0.3
Intention-to-Treat Approach, Censored at 2 y						
KRT	0.49 (0.15-1.63)	1.92 (1.46-2.53)	0.03	1.96 (0.36-10.72)	1.70 (1.14-2.53)	0.9
Overall mortality	0.85 (0.67-1.07)	0.91 (0.79-1.05)	0.6	0.78 (0.59-1.02)	0.97 (0.77-1.22)	0.2
Intention-to-Treat Approach, Censored at 3 y						
KRT	1.10 (0.55-2.21)	1.60 (1.30-1.97)	0.3	0.97 (0.37-2.59)	1.43 (1.04-1.95)	0.5
Overall mortality	0.85 (0.70-1.05)	0.93 (0.82-1.05)	0.5	0.81 (0.63-1.02)	1.02 (0.84-1.24)	0.1
Full Time Analysis						
KRT	1.24 (0.77-1.98)	1.56 (1.31-1.86)	0.4	1.09 (0.44-2.68)	1.33 (1.03-1.71)	0.7
Overall mortality	0.88 (0.73-1.05)	0.93 (0.83-1.04)	0.6	0.87 (0.70-1.08)	0.97 (0.82-1.16)	0.5

Abbreviations: eGFR, estimated glomerular filtration rate; GLP-1RA, glucagon-like peptide-1 receptor agonist; KRT, kidney replacement therapy; NA, not available; SGLT2, sodium/glucose cotransporter 2; UACR, urine albumin-creatinine ratio.
[a]*P* values for interaction.

reabsorption, leading to glucosuria, diuresis, natriuresis, uric acid excretion, and reduced intraglomerular pressure, yielding early hemodynamic benefits.[34] GLP-1RAs primarily reduce weight and cardiovascular risk while also activating local GLP1 receptor–mediated pathways.[35] Both classes exhibit anti-inflammatory, antioxidant, and antifibrotic effects, targeting markers such as transforming growth factor-β, interleukin-6, and tumor necrosis factor-α.[34] The early hemodynamic benefits of SGLT2 inhibitors may enhance their kidney protection, particularly in high-risk populations.

Our study used a comprehensive national health database, capturing extensive data from hospital and outpatient visits, which bolsters the representativeness and real-world applicability of our findings. To ensure valid and detailed baseline assessments, we considered key clinical parameters and applied advanced PS methods, including matching and overlap weighting, to balance patient characteristics. Overlap weighting emphasized patients in treatment equipoise, enhancing balance and minimizing bias while reducing the effective sample size to 10,892, ensuring robust group comparability. Additionally, we adopted an intention-to-treat follow-up approach with multiple censoring durations to track the effects of treatment over time. Together, these methodologies strengthened the robustness of our study, closely emulating key elements of randomized controlled trials and ensuring reliable insights into treatment effects.

Our findings should be interpreted in light of the study's limitations. First, the retrospective observational design limits causal inference and may allow for residual confounding despite robust adjustments. The limited use of GLP-1RAs (approximately 10%) in this cohort reflects barriers such as injection challenges, supply shortages, and Taiwan National Health Insurance Research Database reimbursement criteria prioritizing patients with poorly controlled diabetes (HbA_{1c} >8.5%) or cardiovascular conditions. During the study period, SGLT2 inhibitors were primarily prescribed to patients with preserved kidney function (eGFR ≥45 mL/min/1.73 m^2) because of early safety concerns, whereas GLP-1RAs were favored in advanced CKD for their safety profile. Consequently, GLP-1RA users in the original cohort had a lower eGFR and greater albuminuria, reflecting more advanced CKD. Although PS methods addressed these baseline differences and similar mortality rates suggest reduced bias, residual confounding from unmeasured factors related to prescribing practices remains possible.

Second, the generalizability of our findings may be limited to East Asian individuals with type 2 diabetes who were predominantly nonobese, had poor glycemic control, and had CKD stages G3a-G5. Excluding patients without eGFR data (10%) likely had minimal impact. The study focused on specific drugs—SGLT2 inhibitors (empagliflozin, dapagliflozin) and GLP-1RAs (liraglutide, dulaglutide)—which may restrict applicability to populations with different demographic characteristics, obesity rates, glycemic control, CKD stages, or treatment regimens.

Third, the small sample size in some subgroups restricted the power to detect meaningful differences between SGLT2 inhibitors and GLP-1RAs. Fourth, to enroll more patients with potential kidney function impairment, we determined study inclusion using the most recent eGFR measurements rather than the mean values, and therefore did not consider acute fluctuations in kidney function. Fifth, KRT as the primary end point is limited by variability in clinical decisions and its inability to capture earlier disease progression. A supplementary analysis with a composite end point (KRT or 40% eGFR decrease) showed consistent results, reinforcing the robustness of our findings. Future studies may consider GFR thresholds or kidney failure–related deaths for a more comprehensive assessment. Finally, the database did not include detailed information regarding causes of kidney failure, kidney biopsy results, or specific causes of death. It is presumed that KRT in this cohort was primarily due to diabetes. We also did not explore whether there was any potential difference in cause-specific death between the 2 treatment groups because the cause of death was not consistently recorded on death certificates.

In conclusion, the present results suggest that SGLT2 inhibitors may be more effective than GLP-1RAs for reducing progression to KRT, particularly among patients with higher kidney risk profiles. These findings may inform the choice of therapeutic strategies in managing type 2 diabetes in the setting of CKD, accounting for individual kidney and cardiovascular risk factors. Future research is needed to validate these findings in diverse populations and assess the long-term comparative effectiveness of SGLT2 inhibitors and GLP-1RAs, especially weekly formulations.

Supplementary Material

Supplementary File (PDF)

Figure S1: Study cohort assembly for propensity score–matched cohort.

Figure S2: Distributions of propensity score by study drug before and after 1:1 propensity score matching.

Figure S3: Cumulative incidence curves of kidney replacement therapy by study drug.

Figure S4: Cumulative incidence curves of overall mortality by study drug.

Table S1: Anatomical Therapeutic Chemical classification system codes used to identify use of study drugs.

Table S2: *International Classification of Diseases*, *9th* or *10th Revision*, *Clinical Modification* diagnosis or procedure codes or Taiwan health insurance service claims codes used to exclude patients with high-risk conditions.

Table S3: *International Classification of Diseases*, *9th* or *10th Revision*, *Clinical Modification* diagnosis codes and Taiwan health insurance service claims codes used to identify study outcomes.

Table S4: *International Classification of Diseases*, *9th* or *10th Revision*, *Clinical Modification* diagnosis or procedure codes or Taiwan health insurance service claims codes used to identify smoking status, comorbidities, and measures of health care services at baseline.

Table S5: Anatomical Therapeutic Chemical classification system codes or Taiwan health insurance service claims codes used to identify medication use at baseline.

Table S6: Mean values of clinical parameters before and after propensity score matching.

Table S7: Adherence data: discontinuation and therapy switching rates among GLP-1RA and SGLT2 inhibitor users during the first, second, and third years of follow-up.

Table S8: Baseline demographics, comorbidities, other medication use, and resource use measured within 365 days before the index date before and after propensity score matching using the logit scale with a caliper width of 0.2 times the standard deviation of the logit-transformed propensity score.

Table S9: Follow-up duration and incidence rates of outcomes for GLP-1RA versus SGLT2 inhibitor initiators in original and propensity score–matched cohorts using the logit scale with a caliper width of 0.2 times the standard deviation of the logit-transformed propensity score.

Table S10: Main analysis: hazard ratios and 95% confidence intervals of composite kidney end point (KRT or 40% decrease in eGFR) comparing GLP-1RA initiators with SGLT2 inhibitor initiators.

Table S11: Baseline demographics, comorbidities, other medication use, and resource use measured within 365 days before the index date before and after overlap weighting.

Table S12: Follow-up duration, number of cases, and incidence of outcomes among GLP-1RA and SGLT2 inhibitor initiators before and after overlap weighting.

Table S13: Sensitivity analysis: adjusted hazard ratios and 95% confidence intervals of outcomes comparing overlap-weighted cohort of GLP-1RA initiators with SGLT2 inhibitor initiators excluding those with cancer or autoimmune disease and excluding those with eGFR <15 mL/min/1.73 m^2.

Table S14: Subgroup analysis: hazard ratios and 95% confidence intervals of outcomes comparing overlap-weighted cohort of GLP-1RA initiators with SGLT2 inhibitor initiators among those with an eGFR of 45-59 mL/min/1.73 m^2 and <45 mL/min/1.73 m^2 and those with a UACR ≥300 mg/g and <300 mg/g.

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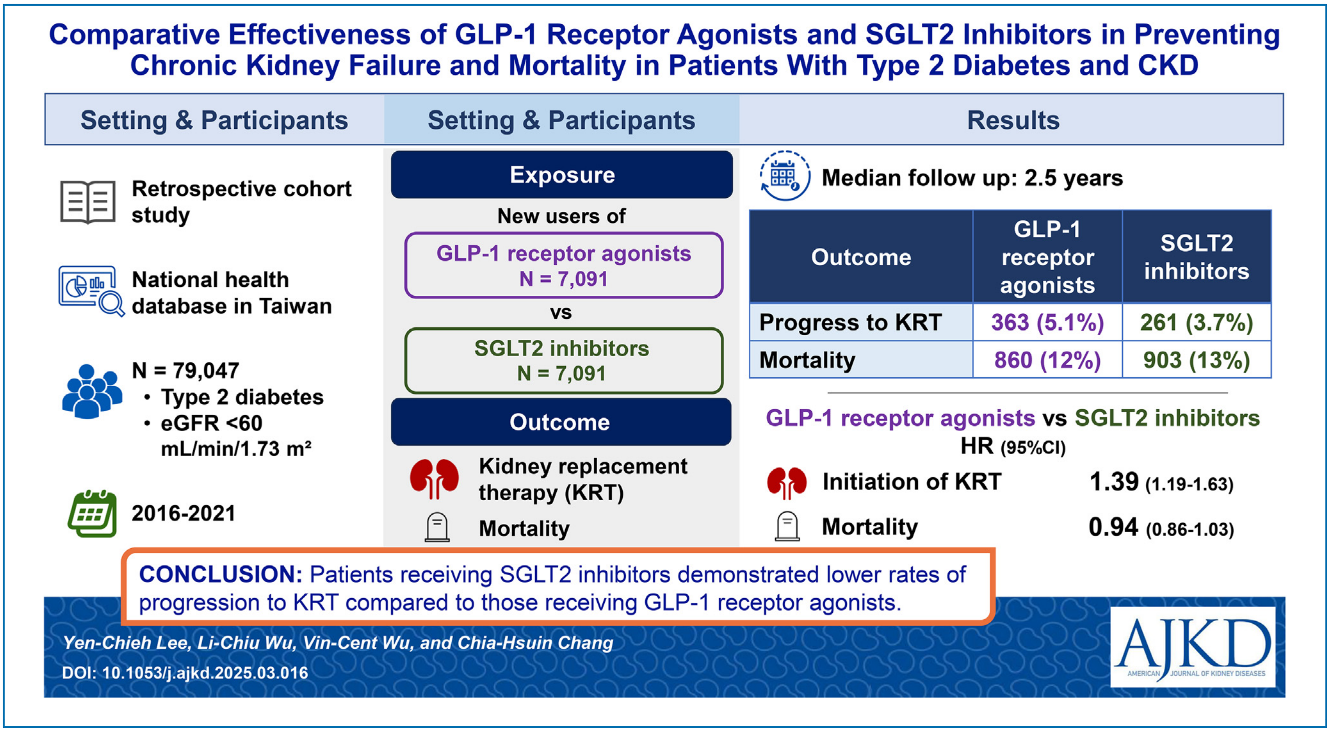
Comparative Effectiveness of GLP-1 Receptor Agonists and SGLT2 Inhibitors in Preventing Chronic Kidney Failure and Mortality in Patients With Type 2 Diabetes and CKD
Setting & Participants
Setting & Participants
Results
Retrospective cohort study
National health database in Taiwan
N = 79,047
• Type 2 diabetes
• eGFR <60 mL/min/1.73 m²
2016-2021
Exposure
New users of
GLP-1 receptor agonists N = 7,091
vs
SGLT2 inhibitors N = 7,091
Outcome
Kidney replacement therapy (KRT)
Mortality
Median follow up: 2.5 years
Outcome | GLP-1 receptor agonists | SGLT2 inhibitors
Progress to KRT | 363 (5.1%) | 261 (3.7%)
Mortality | 860 (12%) | 903 (13%)
GLP-1 receptor agonists vs SGLT2 inhibitors
HR (95%CI)
Initiation of KRT 1.39 (1.19-1.63)
Mortality 0.94 (0.86-1.03)
CONCLUSION: Patients receiving SGLT2 inhibitors demonstrated lower rates of progression to KRT compared to those receiving GLP-1 receptor agonists.
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